



SIMATS Online Programming Lab

C PROGRAMMING



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CO2 - AT2 - SCENARIO BASED QUESTIONS

[← Back](#)

QUESTIONS

Q1

Scenario Based - Smart Waste Recycling Plant System
25 marks

Q2

Scenario Based - IT Company Project Performance Evaluation (2025–2026).
25 marks

Q3

Scenario Based - 3. Semiconductor Chip Manufacturing System
25 marks

Q4

Scenario Based - 4. Cloud Storage Subscription Management (2026)
25 marks

4/4 answered

Q1

Scenario Based - Smart Waste Recycling Plant System

25
marks

A recycling plant processes 5,000 kg of waste every day.
Waste Categories:

Plastic

Paper

Metal

Electronic Waste

Efficiency Levels:

≥90% Recycled → Excellent

75–89% → Good

50–74% → Average

Below 50% → Poor

(a) Develop a C program using arithmetic operators to calculate recycled waste percentage and remaining waste. Use decision-making statements to determine plant efficiency.

(b) Use looping constructs to process waste data for 30 days. Use continue to skip maintenance days and break if the processing machine fails.

Your Code

```
1 #include <stdio.h>
2
```

```
3 int main()
4 {
5     float totalWaste = 5000.0, recycledWaste
6     int maintenance, machineFail;
7
8     scanf("%f", &recycledWaste);
9
10    percentage = (recycledWaste / totalWaste
11
12    printf("Recycled Percentage = %.2f%%\n",
13
14    if (percentage >= 90)
15        printf("Excellent\n");
16    else if (percentage >= 75)
17        printf("Good\n");
18    else if (percentage >= 50)
19        printf("Average\n");
20    else
21        printf("Poor\n");
22
23    for (int day = 1; day <= 30; day++)
24    {
25        scanf("%d", &maintenance);
26
27        if (maintenance == 1)
28            continue;
29
30        scanf("%d", &machineFail);
31
32        if (machineFail == 1)
33            break;
34
35        printf("Day %d: Processed\n", day);
36    }
37
38    return 0;
39 }
```

Input

4500
0
0
0

Output



Run



Save

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